AI Salary Prediction System

An end-to-end machine learning solution for predicting employee compensation using regression models and real-world HR analytics

Project Objective

Real-World HR Analytics

Develop a predictive system that simulates actual compensation planning scenarios for enterprise use cases.

- Assist companies in strategic compensation planning
- Help candidates understand market salary trends
- Provide data-driven insights for hiring decisions

Data Foundation

Source

Real-world dataset from Kaggle with comprehensive employee compensation records

Key Attributes

Experience level, employment type, job role, company size, location, remote work ratio

Scope

Professional and company-related features enabling accurate market analysis

Data Adapter Layer

Data Cleaning & Standardization

Created a robust preprocessing layer that ensures consistency between raw data and machine learning inputs.

- Handled missing values and outliers
- Standardized categorical formats
- Converted experience levels to numerical values
- Ensured data quality for training

Experience Conversion

Transformed categorical experience levels (Entry, Mid, Senior, Executive) into continuous numerical values for regression modeling.

Feature Engineering Pipeline

01

Base Features

Experience level, job role, company size, location, remote work ratio

02

Polynomial Features

Created experience squared to capture nonlinear salary growth patterns

03

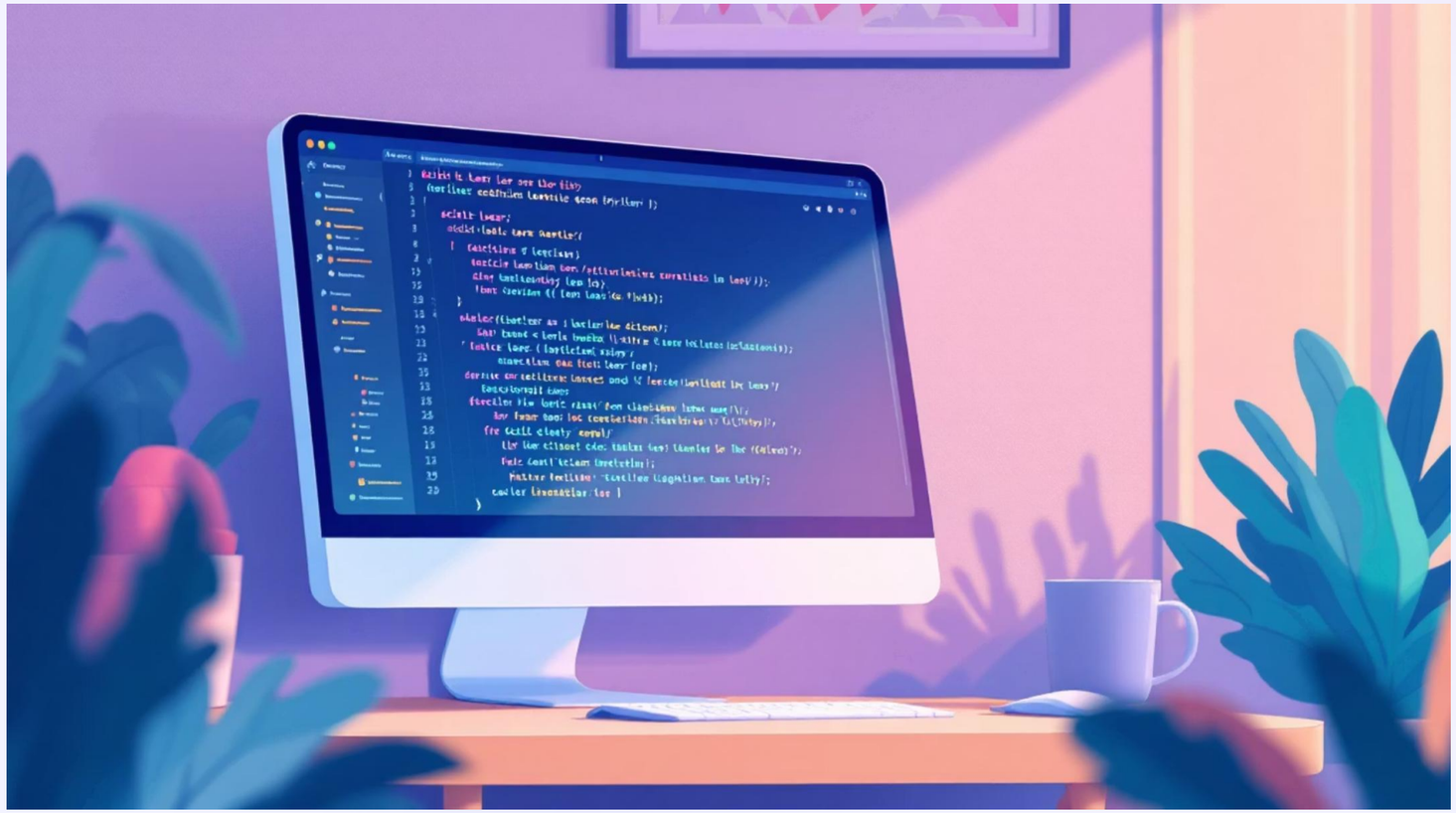
Interaction Terms

Built cross features between skills and experience to model complex relationships

04

Enhanced predictive power through domain-informed transformations

Preprocessing Architecture



Scikit-Learn Pipeline

Built a production-ready ColumnTransformer that applies appropriate transformations based on data type.

Preprocessing pipeline saved with model to guarantee identical transformations during inference.

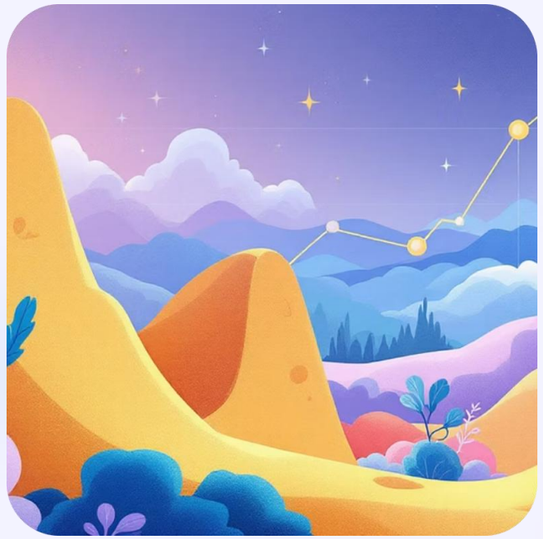
Numerical Features

StandardScaler for normalization

Categorical Variables

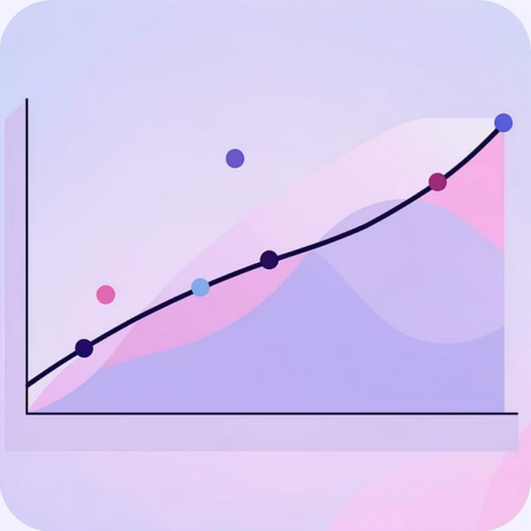
OneHotEncoder for encoding

Model Training & Evaluation



Linear Regression

Baseline model for comparison



Ridge Regression

L2 regularization for overfitting prevention

Random Forest

Ensemble method for robust predictions

XGBoost

Gradient boosting for high performance

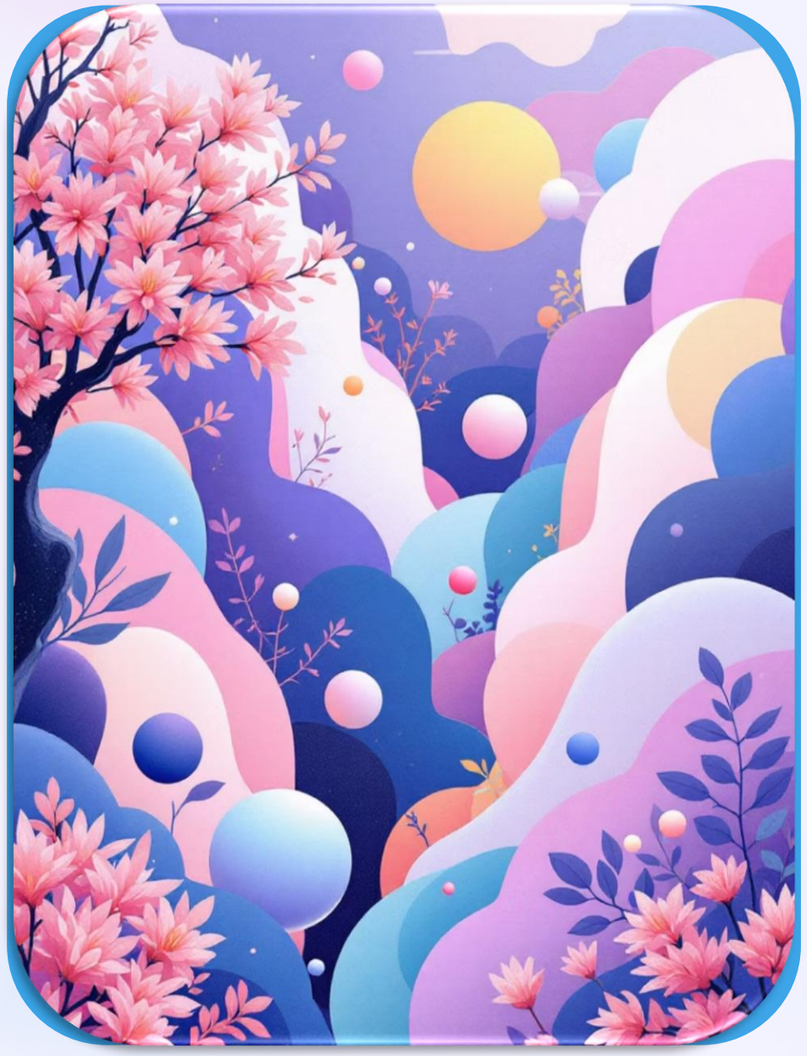
Each model evaluated using R^2 score, with automatic selection of best-performing model saved as deployment-ready artifact.

Explainable AI Integration

SHAP for Transparency

Integrated SHAP (SHapley Additive exPlanations) to remove the black-box nature of machine learning models and provide interpretable insights.

- Identifies which features influence predictions most
- Provides feature importance rankings
- Generates individual prediction explanations
- Builds stakeholder trust through transparency



Streamlit Web Application

User Input

Enter employee details through intuitive interface

Real-Time Processing

Model processes features instantly

Salary Prediction

Receive estimated compensation in milliseconds

Deployed production-ready web application where users can input employee attributes and receive predictions in real time.

Complete ML Lifecycle

1. **Data Processing**

   Cleaning, standardization, feature engineering

2. **Model Training**

   Multiple algorithms with automated selection

3. **Explainability**

   SHAP integration for transparency

4. **Deployment**

   Streamlit application for production use

This project demonstrates production-style ML engineering practices, following a complete machine learning lifecycle from data processing through deployment with explainability at each stage.